Oil-Fired Generation

Distillate and Residual Fuel Peakers

Design Basis Accident — Military Action

DBA-ES-GN-FC7 — Facility Class Document

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Preface

The oil-fired peaker is the least glamorous resource in the electric generation portfolio. It burns expensive fuel. It runs rarely — perhaps one to two percent of the hours in a year. Its heat rate is poor, and its emissions per megawatt-hour are high. Every economic and environmental signal points toward its retirement. Yet it persists, stubbornly, in the generation mix of every region that has experienced a serious supply emergency — because when everything else has failed, the oil-fired peaker is what keeps the lights on.

This document establishes the Design Basis Accident scenarios for deliberate military action against oil-fired generating facilities that meet the FC7 criticality threshold. The defining analytical characteristic of FC7 is the inverse relationship between utilization and strategic value: the oil-fired peaker is most important precisely when it is least visible — during grid emergencies, gas supply disruptions, and extended cold-weather demand peaks when every other generation resource has already been committed. An adversary who destroys the peaker fleet destroys the emergency reserve without disrupting normal operations. The attack is invisible in routine performance data until the emergency that requires the peakers arrives, and by then it is too late.

The oil-fired generation fleet is disproportionately concentrated in the Northeast — particularly New England — where it serves as the backstop when gas supply to the gas-dominated generation fleet is curtailed during winter heating season. The gas-electric interdependency in New England, documented in DBA-ES-GC-FC4 and DBA-GAS-FC1, makes oil-fired peakers the final line of grid defense during the compound event: simultaneous gas supply curtailment and high heating demand. The adversary who understands New England grid operations understands that destroying the oil-fired peaker fleet before inducing a gas supply disruption eliminates the backstop before it is needed.

FC7 also introduces the fuel inventory depletion attack as a distinct DBA-MA scenario: an adversary who compromises the automated dispatch system controlling oil-fired peakers can cause spurious starts that burn down the on-site fuel tank before the actual emergency. Unlike coal supply interdiction in DBA-ES-GN-FC6, which requires physical action against the railroad network, oil-fired peaker fuel depletion can be achieved through the automated control system alone — no physical access to the fuel tank is required. The tank empties through the combustion stack, one spurious dispatch at a time, until the emergency arrives and the peaker has nothing left to burn.

1. Introduction

1.1 Purpose

This document establishes the Design Basis Accident — Military Action scenarios for the oil-fired generation facility class. Its purpose is to define the bounding threat event that oil-fired peaker security planning, resilience investments, and recovery architecture must address. As with all DBA-MA documents, this document does not prescribe specific protective measures at individual facilities. It defines the consequence envelope within which protective measures must perform.

An oil-fired peaker security program designed against conventional physical threats will fail when a sophisticated adversary combines automated dispatch system compromise — causing spurious starts that deplete on-site fuel inventory over weeks — with selective kinetic action against the turbines and fuel systems at the highest-consequence peaker facilities, timed to execute immediately before a gas supply disruption that would otherwise activate the peaker fleet. The DBA-MA approach defines this pre-emptive, backstop-removal attack as the governing design event.

1.2 Scope and Applicability

This document addresses oil-fired generating facilities meeting either of two applicability criteria: individual generating unit nameplate capacity at or above 100 MW burning distillate or residual fuel oil as the primary fuel, or designation as a reliability must-run (RMR) unit by the applicable RTO or ISO, regardless of nameplate rating. The 100 MW threshold for FC7 is lower than the 750 MW threshold for FC6, reflecting the peaker fleet's role as emergency reserve rather than baseload generation: a 100 MW oil-fired peaker in a transmission-constrained emergency context can be the difference between controlled demand response and involuntary load shedding. Must-run designation triggers FC7 applicability for any oil-fired unit, regardless of size, where the RTO has determined the unit is necessary for system reliability.

The facility class scope encompasses combustion turbine peakers burning distillate or residual fuel oil, older oil-fired steam units, on-site fuel storage tanks and transfer systems, automated start and dispatch control systems, and remote operations infrastructure. Dual-fuel units that can burn either natural gas or distillate are within scope when operating in oil-fuel mode or when their oil-fuel capability is the basis for the RMR designation. Units that are nominally dual-fuel but have not demonstrated oil-fuel capability through recent commissioning tests are treated as gas-only units and are outside FC7 scope.

1.3 Relationship to Governing Documents

This document is issued under the DBA-ES Electric Sector Framework (v0.7), which is itself issued under the DBA-EN-SF Energy Sector Framework (v0.3) and the DBA-MA Parent Framework (IAEA L1 v1.4). The GN type code and FC numbering convention are established in DBA-ES-GN-FC6. The threat taxonomy, bounding methodology, consequence analysis structure, and adequacy standards are inherited from the sector framework. The RT_HVLLC metric is inherited from the DBA-ES series and calibrated to oil-fired generation equipment specifications in Section 6.1.

The gas-electric interdependency consequence chain documented in DBA-ES-GC-FC4 and DBA-GAS-FC1 is the primary cross-series reference for FC7: the oil-fired peaker fleet is the electric sector's response to gas supply curtailment, and its destruction or fuel depletion before a gas supply disruption produces the compound consequence that is the governing scenario for this facility class. The Dual Design Basis architecture and Bright Line / Command Broker requirements are inherited unchanged and applied to the peaker automated dispatch environment in Section 3.3.

1.4 The Last-Resort Generator — Strategic Value Inversely Proportional to Utilization

No facility class in the DBA-ES series presents a greater mismatch between operational visibility and strategic importance than the oil-fired peaker. Its annual capacity factor is typically 1 to 3 percent — it runs perhaps 90 to 260 hours per year. It does not appear in routine generation dispatch data at a level that attracts operational attention. Its fuel costs make it uncompetitive in normal market conditions. Its emissions profile makes it politically uncomfortable to defend. Yet in every major Northeast grid emergency of the past two decades — the ice storms, the polar vortex events, the natural gas curtailments — oil-fired peakers provided the margin between controlled emergency operations and uncontrolled cascading failure.

This invisibility is the adversary's primary operational advantage. A physical security vulnerability at a coal plant or transmission substation is visible in the CIP-014 vulnerability assessment process, triggers regulatory attention, and generates investment in hardening. A physical security vulnerability at an oil-fired peaker that runs 150 hours per year may never be assessed against the DBA-MA threat tier because no regulatory framework requires it — and because the facility's low utilization creates an institutional tendency to underestimate its strategic importance. The DBA-MA Design Basis is precisely the tool for correcting that underestimation.

2. Facility Profile and Asset Taxonomy

2.1 Combustion Turbine Peakers

The dominant technology in the modern oil-fired peaker fleet is the simple-cycle combustion turbine — a gas turbine adapted to burn distillate fuel oil (No. 2 fuel oil, chemically similar to diesel) when natural gas is unavailable or uneconomic. Simple-cycle combustion turbines have fast start capabilities — many can reach full load within 10 to 30 minutes of a start command — that make them well-suited for emergency response applications. Their thermal efficiency is low compared to combined-cycle gas plants, but efficiency is not the relevant metric for a unit whose function is emergency capacity rather than economic dispatch.

Dual-fuel combustion turbines — designed to burn either natural gas or distillate — are the preferred peaker configuration in regions where both fuels are available but where gas supply reliability is uncertain. In New England, where gas pipeline congestion during the winter heating season can curtail gas-fired generation, dual-fuel capability is a reliability requirement for RMR units. The distillate fuel system — fuel oil storage tanks, transfer pumps, day tanks, and fuel forwarding systems — is an integral component of a dual-fuel combustion turbine's reliability designation.

2.2 Older Oil-Fired Steam Units

A subset of the oil-fired fleet consists of older steam-generating units originally built as baseload or intermediate units and retained as peakers after economic conditions made frequent dispatch uneconomic. These units burn residual fuel oil (No. 6 fuel oil) — a heavier, higher-viscosity product that requires preheating before combustion and has a more specialized supply chain than distillate. Residual fuel oil is a byproduct of the refining process; its supply is concentrated at refineries and marine terminals and is transported by tanker or barge rather than through the distillate distribution system.

Older steam units have higher RT_HVLLC values than combustion turbines — steam turbine components with lead times of 12 to 20 months for major repairs — but their operational complexity also makes them more vulnerable to extended outage from DCS compromise. The boiler startup sequence for a cold residual-fuel oil steam unit requires precise management of fuel oil temperature, burner management system sequencing, and steam drum level control; an adversary who has compromised the DCS can prevent a successful start without causing any physical damage.

2.3 Fuel Types and On-Site Storage

Distillate fuel oil (No. 2). The primary fuel for modern combustion turbine peakers. Chemically similar to highway diesel; distributed through the same terminal and pipeline infrastructure as heating oil and diesel fuel. Distillate is stored in on-site tanks — typically 30,000 to 300,000 gallons per unit, representing 30 to 90 days of full-load operating fuel depending on tank size and unit heat rate. Deliveries are by tanker truck or, at larger facilities, by barge. The distillate supply chain has multiple alternative routing options — tanker truck delivery to remote sites, barge delivery to waterfront sites — making it more resilient to single-point interdiction than coal rail supply. However, on-site storage is the sole buffer: there is no continuous delivery during operation, and the tank empties as the unit runs.

Residual fuel oil (No. 6). The fuel for older steam peakers. Heavier, more viscous, and requiring heating before combustion. The residual fuel supply chain is more specialized and less geographically distributed than the distillate: major suppliers are concentrated at refineries and marine terminals with deep-water access for tanker delivery. On-site storage for residual fuel is typically larger than for distillate — 90 to 180 days of full-load fuel — reflecting both the longer lead time for supply delivery and the historical expectation of extended seasonal operation. Residual fuel oil supply from the same refinery and terminal network addressed in DBA-OIL-FC1 and DBA-OIL-FC3.

2.4 Remote Operations Architecture

Oil-fired peakers are among the most remotely operated generating assets in the U.S. fleet. Many facilities are unmanned except during startup and maintenance periods; routine operation, monitoring, and dispatch response are managed from a remote operations center. The automated start sequence — initiated by a remote dispatch signal from the RTO or the plant operator — manages fuel system priming, inlet guide vane positioning, ignition, synchronization, and ramp to full load without on-site personnel.

This remote operations architecture creates an attack surface unique in the DBA-ES generation series. A coal plant or large gas turbine operates with continuous on-site staffing; an adversary who has compromised the DCS must contend with operators who are physically present and observing process conditions. An oil-fired peaker with remote-only operation has no on-site observer during normal conditions. A spurious start sequence initiated by a compromised automated dispatch system may run its full cycle — consuming fuel and accumulating operating hours on the combustion turbine — before a remote operator detects the anomaly.

2.5 Northeast Peaker Concentration Profile

Oil-fired generation is geographically concentrated in the Northeast — ISO-NE and PJM Interconnection — where it accounts for a disproportionate share of emergency generation capacity relative to its share of total installed capacity. ISO-NE operates under a capacity market that explicitly values oil-fired peakers for their winter fuel security contribution; RMR designations in ISO-NE are disproportionately held by oil-fired units whose distillate storage provides fuel supply assurance during gas pipeline congestion events.

The Northeast peaker concentration means that the FC7 consequence analysis is geographically specific in a way that FC6 is not. A coal plant attack in the Midwest produces grid consequences in the Midwest. An oil-fired peaker attack in New England produces grid consequences specifically during winter gas supply emergencies, which are themselves a New England-specific vulnerability. The Scenario C compound event in this document is therefore a New England-focused analysis, not a national analysis.

2.6 Regulatory Authority

NERC CIP. Oil-fired peakers meeting the BES definition are subject to NERC CIP cybersecurity standards. RMR-designated units are likely to trigger CIP-014 physical security coverage. However, many oil-fired peakers — particularly older units at or below 100 MW that nonetheless hold RMR designations — may fall outside CIP applicability thresholds that were not designed with the peaker fleet's emergency function in mind.

ISO/RTO. The applicable RTO or ISO — ISO-NE, NYISO, PJM — has the primary operational relationship with oil-fired peakers through capacity market obligations, RMR agreements, and dispatch protocols. The RTO's knowledge of which units are fuel-secure (have verified on-site distillate inventory) versus nominally dual-fuel but operationally gas-dependent is the most consequential single piece of intelligence for emergency planning — and it is information that an adversary with access to capacity market filings can obtain from public sources.

EPA. Oil-fired generation is subject to EPA air quality regulations under the Clean Air Act. The National Emission Standards for Hazardous Air Pollutants (NESHAP) for stationary combustion turbines and the emission limits for distillate and residual fuel combustion affect the permissible operating hours for some peaker units. Emissions permit constraints that limit annual operating hours create a perverse vulnerability: a unit that has been forced to run by spurious automated dispatch may exhaust its annual permitted hours before the actual emergency that requires it.

3. Design Basis: Oil-Fired Generation Facility Class

3.1 Non-Adversarial Bounding Events

The non-adversarial Design Basis for the oil-fired generation facility class inherits the infrastructure failure event, operational event, and natural hazard event categories from the DBA-ES Sector Framework. This document develops the facility-class-specific consequence chains for three governing non-adversarial bounding events.

Fuel tank fire or containment failure. The governing process safety bounding event is a major fuel oil tank fire or secondary containment failure producing a significant fuel oil release. Distillate fuel oil storage tanks at peaker facilities are subject to the same overfill and static electricity ignition risks as petroleum product storage tanks generally; the DBA-OIL-FC3 tank farm fire bounding event applies at a smaller scale. The governing non-adversarial case involves a tank fire during a cold-weather emergency when the facility is operating at high utilization — consuming its fuel supply faster than normal and leaving less inventory buffer against the fire-related loss.

Automated start failure during emergency dispatch. Failure of the automated start sequence during an emergency dispatch event — a cold start failure due to fuel system priming issues, inlet guide vane malfunction, or DCS software fault — leaving the grid operator without the expected peaking capacity during an emergency. Automated start reliability is the primary operational concern for peaker fleet management; RTOs track oil-fired peaker start reliability statistics and include start failure probabilities in emergency operating procedure planning.

Fuel supply delivery failure during extended cold weather. Extended cold-weather operation at high utilization rates depletes on-site fuel inventory faster than normal delivery schedules can replenish it. During the 2018 polar vortex events, several New England oil-fired peakers operated continuously for days — consuming fuel at rates that challenged normal delivery logistics. The non-adversarial bounding event involves a fuel delivery failure during extended high-utilization operation, producing fuel exhaustion and forced shutdown during ongoing grid emergency conditions.

3.2 Design Basis Accident — Military Action: Bounding Event

The DBA-MA bounding event for the oil-fired generation facility class is a coordinated attack combining automated dispatch system compromise — causing spurious starts that deplete on-site fuel inventory over a 3 to 5 week horizon — with precision kinetic action against the combustion turbine combustion sections and fuel systems at the highest-consequence RMR peaker facilities, executed immediately before a gas supply disruption that would normally activate the oil-fired peaker fleet. The attack removes the backstop before the crisis that requires it.

The bounding parameters, inherited from the DBA-ES Sector Framework threat taxonomy:

Adversary tier: Tier 1 state actor with automated dispatch system pre-positioning capability. The remote operations architecture of oil-fired peakers reduces the operational threshold for the fuel depletion component: SCADA access to the dispatch system may be achievable through the remote operations center's IT-OT boundary without requiring a separate intrusion into on-site plant control systems.

Modality: Cyber-primary with kinetic amplification. The cyber component — automated dispatch system compromise causing spurious starts — depletes on-site fuel inventory over 3 to 5 weeks, with each spurious dispatch appearing as a normal test start or market response event in the operations logs. Kinetic action against the combustion section and fuel forwarding system executes against the highest-consequence RMR units at the moment the gas supply disruption begins, ensuring that even units with remaining fuel cannot start due to physical damage.

Target selection: RMR-designated oil-fired peakers in New England (ISO-NE) and Mid-Atlantic (PJM) whose combined capacity represents the grid's margin above involuntary load shedding during the governing winter emergency scenario. The adversary selects the minimum set of units whose combined loss crosses the threshold from controlled emergency operations to uncontrolled load shedding.

Timing: Immediately before a gas supply disruption that the adversary has either induced (DBA-GAS-FC1 Scenario A) or is anticipated on the basis of weather forecasting — a major cold snap that will trigger high heating demand and gas pipeline congestion in New England. The adversary exploits the same weather forecast visibility available to the grid operator: the 7 to 10 day forecast that identifies the coming demand peak is the same forecast that tells the adversary when the peakers will be most needed.

Simultaneity: Coordinated attacks on multiple RMR peaker facilities within ISO-NE within a 48-hour window — exhausting the ISO-NE emergency capacity margin simultaneously across multiple sites.

3.3 Dual Design Basis Architecture

Consistent with the DBA-ES Sector Framework and DBA-EN-SF (§4.2), an oil-fired generating facility may simultaneously carry a sector-specific physical Design Basis established by this document and an AI Governance Design Basis established under the AI Governance series. The two design bases interact at the DBA level. The Bright Line and Command Broker architecture established in CB-Framework_Cross_Sector v0.1 governs the boundary between the two design bases.

Three oil-fired-generation-specific AI governance interaction points are identified for the FC7 Design Basis:

Automated start and dispatch control systems. Automated start sequencing and dispatch response systems that initiate combustion turbine starts in response to RTO dispatch signals or pre-programmed test schedules are ML systems operating below the Bright Line when performing deterministic sequencing functions within their designed operating envelopes. They may operate autonomously for programmed test starts and dispatch a response within defined parameters. However, automated dispatch is precisely the attack surface exploited in Scenario B: a compromised automated dispatch system that initiates spurious starts — each individually appearing as a normal test or market response event — depletes fuel inventory without any operator intervention. The AI Governance Design Basis must establish that automated dispatch systems log all start events with a verified dispatch authorization identifier, that cumulative operating hours in any rolling 30-day window are compared against the seasonal dispatch plan with anomalous deviations triggering human review, and that no automated start sequence can execute without a verified dispatch authorization signal from the RTO or plant operator — a simple authorization check that prevents spurious starts initiated by a compromised local automation system.

Fuel inventory management and delivery scheduling ML. ML systems that monitor on-site fuel tank levels, project fuel consumption based on dispatch forecasts, and schedule tanker truck or barge deliveries are below the Bright Line when performing deterministic inventory management functions. The adversarial concern is manipulation of inventory reporting — causing the ML system to report higher tank levels than actually exist, masking the depletion caused by spurious automated starts. The Design Basis requirement is that fuel inventory reporting systems receive tank level data from independent sensors that are architecturally separate from the automated dispatch system, and that inventory readings that decline faster than the dispatch log can explain trigger an immediate investigation regardless of the inventory management ML system's reported status.

Remote operations center Gen AI advisory tools. Gen AI advisory tools for remote peaker fleet operations — tools that recommend dispatch sequencing, maintenance scheduling, and emergency response prioritization across a portfolio of oil-fired peakers — are above the Bright Line and require Command Broker authorization. In the emergency context that defines the peaker fleet's strategic function, the Command Broker requirement is most consequential: a Gen AI tool whose emergency dispatch recommendations are accepted without human review during a grid emergency can misdirect the available peaker capacity in ways that maximize the adversary's intended consequence. The Command Broker for emergency dispatch Gen AI tools must be the qualified system operations engineer with authority over the peaker fleet dispatch, not the Gen AI system itself.

4. Design Basis Accident Scenarios

4.1 Scenario A: Pre-Emptive Destruction of Peaker Fleet Before Induced Gas Supply Disruption

4.1.1 Initiating Event

Coordinated kinetic attacks against the combustion turbine hot sections and fuel forwarding systems at four to six major RMR-designated oil-fired peaker facilities in New England, executed during a 48-hour window immediately preceding the initiation of a gas supply disruption affecting New England gas-fired generation. The kinetic attacks target the combustion liner and transition piece assemblies — the high-temperature components whose replacement requires 8 to 12 weeks — and the fuel forwarding pumps and day tanks that supply distillate to the combustion turbine. The fuel system attacks are designed to ensure that even units with full on-site tank inventory cannot start: the combustion section damage prevents operation, and the fuel system damage adds a redundant denial pathway.

The gas supply disruption initiates 24 to 48 hours after the peaker attacks — either through a coordinated DBA-GAS-FC1 compressor station attack or through a weather-driven pipeline congestion event that the adversary has timed to maximize. The ISO-NE operator, facing both curtailed gas-fired generation and unavailable oil-fired peakers, must declare an Energy Emergency Alert (EEA) at a level requiring involuntary load shedding significantly earlier than the available demand response and import capacity can prevent.

4.1.2 Progression Sequence

T-48 to T+0 hours: Kinetic strikes against four to six New England RMR peaker facilities. Each attack is a precision strike against an unmanned or lightly-staffed remote facility — detectable but not immediately attributable as a coordinated pattern by the individual plant operators. ISO-

NE operational reliability team receives outage notifications for the affected units; initial attribution to equipment failures, not a coordinated attack.

T+0 to T+24 hours: Gas supply curtailment begins as weather-driven demand increases, resulting in pipeline throughput constraints. ISO-NE issues EEA Level 1. Gas-fired generation is ramping down in sequence as fuel supply pressure falls. ISO-NE dispatches remaining oil-fired peakers — units not affected by the kinetic attack respond. The gap between expected peaker capacity (including attacked units) and available peaker capacity becomes visible to the ISO-NE operator as the emergency deepens.

T+24 to T+72 hours: Gas curtailment at maximum. ISO-NE at EEA Level 2 or 3. Available peaker capacity is insufficient to maintain the load without involuntary load shedding. ISO-NE activates controlled load shedding across New England. The coordinated nature of the peaker attacks is confirmed by the FBI and CISA investigation — physical damage patterns inconsistent with equipment failure.

T+72 hours onward: Load shedding continuing until gas supply is restored or demand subsides. Peaker repair assessment: combustion liner and transition piece replacement, 8 to 12 weeks per unit. Fuel system repair, 2 to 4 weeks per unit. All affected units are offline for the duration. DOE ESF-12 and FEMA mass care coordination for the affected New England population.

4.1.3 Consequence Envelope

Involuntary load shedding duration. The governing consequence is the duration of controlled load shedding in New England — a densely populated, cold-weather region with limited alternative heating options for the residential population losing electricity. Load shedding during a cold snap produces cascading life-safety consequences: electric heating fails, electric-backup heating fails, water pipes freeze, and vulnerable populations face the same cold-exposure risk identified in DBA-GAS-FC4 for gas distribution outages. The compound consequence of simultaneous gas and electric supply disruption is the worst-case winter residential life-safety event in the DBA-ES series.

RT_HVLLC for combustion turbines. Combustion turbine hot section replacement — combustion liner and transition piece — has a lead time of 8 to 12 weeks under normal conditions for standard frames. Custom components for older turbine frames may have longer lead times. The RT_HVLLC for FC7 is significantly shorter than for FC6, reflecting the more standardized and commercially available nature of combustion turbine components compared to coal steam turbine rotors. However, for simultaneous attacks on four to six units requiring the same components from the same OEM, the effective lead time extends due to queue competition — the same constraint identified for pipe-lay vessels in DBA-OIL-FC2 and specialty contractors in DBA-OIL-FC1.

Emissions permit consequence. Combustion turbines that are repaired and returned to service after an attack must resume operation within their existing air quality permits. If the attack and repair cycle produces operating hours that push the unit toward its annual permitted hours limit, the operator may face a constraint on how much emergency operation the repaired unit can provide in the remainder of the compliance year.

4.2 Scenario B: Fuel Inventory Depletion Through Spurious Automated Dispatch

4.2.1 Initiating Event

Cyber intrusion into the automated dispatch and remote operations systems of multiple oil-fired peaker facilities in New England, with pre-positioned access to the automated start sequencing systems at each facility. The adversary initiates a pattern of spurious automated starts — each appearing in the operations log as either a scheduled weekly test start or a short-duration dispatch response to a market signal — that consumes distillate fuel at a rate slightly above the expected test and dispatch utilization. Over 4 weeks, the spurious dispatch pattern depletes on-site fuel tanks from normal inventory levels of 60 to 90 days down to 5 to 10 days of full-load fuel.

4.2.2 The Invisible Depletion

The Scenario B attack is the oil-fired peaker's analog of the underground storage inventory depletion attack in DBA-GAS-FC3. Both exploit the same temporal asymmetry: the resource depletes quickly (fuel burns fast; gas withdraws fast) and replenishes slowly (tanker truck deliveries require scheduling lead time; injection is compressor-limited). Both produce their consequence in the gap between when the depletion is discovered and when the emergency arrives.

The fuel depletion attack has one additional feature that makes it particularly difficult to detect: each spurious dispatch is individually indistinguishable from a legitimate test or market response event. The ISO-NE capacity market requires oil-fired RMR units to demonstrate start reliability through periodic test starts; an adversary with access to the dispatch system can mimic the timing and duration of legitimate test starts precisely. The aggregate fuel consumption pattern — more starts than scheduled, slightly more fuel consumed than expected — is visible only in a systematic audit of the fuel delivery and consumption records against the documented dispatch log. This review may not occur with sufficient frequency to catch a 4-week depletion campaign.

4.2.3 Consequence Envelope: Fuel Depletion

Fuel delivery restoration timeline. Emergency distillate delivery by tanker truck: 24 to 72 hours from order to delivery for a single facility, longer for simultaneous deliveries to multiple facilities in the same region. The New England distillate distribution system has limited surge capacity — the same terminal and trucking infrastructure serves residential heating oil and highway diesel demand. A simultaneous emergency order for distillate delivery to multiple peaker facilities during peak heating season — when the same trucks are serving residential oil deliveries — may not be fulfillable within the timeline required.

Detection and attribution timing. The fuel depletion attack produces its consequences days to weeks after the adversarial manipulation has been completed. Attribution — confirming that the fuel depletion resulted from spurious automated dispatch rather than legitimate operations — may require a full audit of the dispatch authorization records, which takes days. The grid operator who does not know the peakers are fuel-depleted cannot plan around the gap until the units fail to respond to emergency dispatch. The consequence of the detection delay is that the grid emergency arrives before the emergency fuel delivery has been completed.

Emissions permit interaction. If the spurious automated starts have consumed a significant portion of the unit's annual permitted operating hours, the remaining permitted hours may be insufficient for extended emergency operation even after fuel is restored. The adversary who understands the emissions permit framework can calibrate the spurious dispatch to consume permitted hours as well as fuel. This double constraint limits the unit's emergency response capability through both the physical inventory and the regulatory envelope simultaneously.

4.3 Scenario C: Northeast Peaker Cluster Attack During Winter Storm Uri Analog

4.3.1 Initiating Event

A compound attack executed during a multi-day New England cold snap — outdoor temperatures forecast below 0°F for a sustained period — that simultaneously initiates a gas pipeline compressor station attack affecting New England gas supply (per DBA-GAS-FC1 Scenario A) and executes Scenario A kinetic strikes against the New England oil-fired peaker fleet. The compound event replicates the Winter Storm Uri consequence envelope — simultaneous gas supply disruption and generation loss — but with the key difference that the Uri event was non-adversarial and produced recoverable operational decisions, while Scenario C involves physical destruction of both the gas supply infrastructure and the emergency generation backstop.

4.3.2 The Uri Analog Applied

Winter Storm Uri (February 2021) demonstrated the gas-electric interdependency in extreme cold-weather conditions at a national scale: gas supply curtailment forced gas-fired generation offline; demand for both gas heating and electric heating exceeded available supply; the compound failure produced load shedding across multiple states and directly contributed to hundreds of deaths. The New England version of this event is structurally similar but geographically more contained and more dependent on oil-fired backup: ISO-NE's emergency operating procedures assume oil-fired peaker availability as the backstop for gas curtailment.

Scenario C removes that backstop before the crisis. With both gas supply curtailed (through the DBA-GAS-FC1 component) and oil-fired peakers physically destroyed or fuel-depleted (through the Scenario A or B component), ISO-NE has no remaining dispatchable generation margin above load shedding. The demand response resources, import capability, and conservation voltage reduction that can defer load shedding in a single-fuel emergency cannot compensate for the simultaneous loss of both emergency supply pathways.

4.3.3 Consequence Envelope: Compound Winter Emergency

Extended load shedding with cold-weather life-safety. Multi-day load shedding in New England during sub-zero outdoor temperatures is the highest-consequence residential life-safety event in the DBA-ES series for the Northeast region. The combination of gas heating failure (gas supply curtailed) and electric heating failure (load shedding) eliminates both primary and backup heating for residential customers simultaneously. The consequence profile is similar to the DBA-GAS-FC4 Scenario C outcome, but with the electric dimension added: customers who would have maintained electric backup heating during a gas-only outage lose both simultaneously.

Military logistics consequence. New England military installations — naval facilities at Groton and Portsmouth, Coast Guard operations, and Air Force installations — depend on the same civilian electric grid during peacetime operations. Extended load shedding that affects military base power supply degrades military readiness at a moment when the adversary has presumably created the compound crisis as a component of a broader military action context. Military backup generation systems are sized for hours to days, not the multi-day outage that Scenario C produces.

Cross-sector cascade. Electric service loss in a cold-weather emergency triggers a cascade across every sector dependent on electricity: water treatment and pumping, wastewater treatment, telecommunications, hospital and emergency services operations, and transportation. The compound gas-electric failure eliminates the ability to maintain even reduced operations in sectors that would normally be able to continue under single-fuel disruption conditions.

5. Protective Action Thresholds

5.1 Threat Detection and Indicator Taxonomy

5.1.1 Elevated Threat Indicators

Intelligence Community reporting of adversary interest in New England grid emergency procedures or oil-fired peaker fleet operational details. Anomalous automated dispatch system activity: start events that are not corroborated by RTO dispatch records or scheduled test authorizations; fuel consumption rates inconsistent with the documented dispatch log. Aggregate on-site fuel inventory is declining below seasonal norms across multiple peaker facilities within the same RTO footprint without a documented dispatch justification. Physical surveillance of peaker facility sites or remote operations center locations. Weather forecast analysis indicating an upcoming demand peak that would trigger peaker dispatch — an adversary who knows when the peakers will be needed can work backward to the pre-attack window.

5.1.2 Imminent Threat Indicators

Confirmed unauthorized access to peaker automated dispatch or remote operations systems. Fuel inventory at two or more peaker facilities is simultaneously at critically low levels without a documented dispatch explanation. Confirmed unauthorized UAS or ground access at any RMR peaker facility. Simultaneous equipment anomalies at multiple peaker sites within the same RTO footprint within 48 hours. Confirmed kinetic event at any oil-fired peaker facility. ISO-NE operational reliability alert identifying multiple RMR units simultaneously unavailable during a forecast peak demand period.

5.2 Decision Authority Framework

Plant operator and remote operations center: The remote operations supervisor has the authority to halt automated dispatch sequences, declare units unavailable, and initiate emergency fuel delivery orders. NERC CIP incident reporting for confirmed or suspected cyber incidents affecting automated dispatch systems. ISO-NE notification for any RMR unit unavailability. EPA notification for fuel oil releases. FBI and CISA notification for confirmed or suspected adversarial automated dispatch manipulation.

RTO/ISO level: ISO-NE emergency operating procedures for loss of multiple RMR units: EEA declaration sequence, import maximization, demand response activation, and load shedding authorization. NERC event analysis notification. DOE notification for grid emergencies triggering ESF-12.

Federal coordination: DOE ESF-12 for regional grid emergency response. CISA for automated dispatch system cyber incident coordination. FEMA for mass care coordination during load shedding events with cold-weather life-safety consequences. FBI and DOD for events with confirmed or strongly suspected state-actor attribution. HHS for cold-weather medical consequence coordination.

5.3 Federal Coordination Thresholds

Confirmed kinetic attack on any RMR-designated oil-fired peaker facility. Confirmed automated dispatch system compromise with suspected nation-state attribution. Aggregate fuel inventory at RMR oil-fired peakers in a single RTO footprint falling below a defined emergency threshold during a forecast peak demand period. Simultaneous unavailability of three or more RMR peaker units in the same RTO footprint during a grid emergency. ISO-NE EEA Level 2 or higher, with oil-fired peaker unavailability as a contributing factor. Compound gas-electric emergency conditions during a cold-weather event.

6. Recovery Architecture

6.1 RT_HVLLC Calibration for Oil-Fired Generation

The RT_HVLLC for FC7 is significantly shorter than for FC6, reflecting the more standardized and commercially available nature of combustion turbine components compared to coal steam turbine rotors and supercritical boiler pressure parts. The governing RT_HVLLC values for the two primary FC7 equipment types:

Combustion turbine hot section (combustion liner, transition piece, first-stage turbine nozzle): 8 to 14 weeks for standard frame sizes from current OEM production inventory; 16 to 24 weeks for older or uncommon frame sizes requiring custom fabrication or refurbishment. The effective RT_HVLLC extends when simultaneous attacks on multiple units create queue competition at the OEM, as discussed in Section 4.1.3.

Combustion turbine fuel forwarding system (pumps, day tanks, fuel filters): 2 to 6 weeks — off-the-shelf components for most combustion turbine fuel systems. Not the governing RT_HVLLC constraint unless the attack has also damaged the turbine.

Older oil-fired steam turbine major components: 12 to 20 months for major turbine section repairs — approaching the FC6 range for the subset of the fleet consisting of older steam units. The governing RT_HVLLC for the steam unit subset of FC7 is therefore in the DBA-ES-TX-FC2 LPT range rather than the shorter combustion turbine range.

6.2 Recovery Sequencing

Phase 1 — Immediate (0–72 hours). Emergency fuel delivery orders to all affected facilities with remaining fuel capability. Automated dispatch system forensic investigation initiation. Grid operator emergency operating procedures for RMR unit unavailability. ISO-NE EEA declaration and emergency import maximization. Physical damage assessment at kinetically attacked facilities. DOE ESF-12 activation for regional grid emergency. CISA notification for the dispatch system cyber component.

Phase 2 — Short-term (72 hours to 6 weeks). Emergency fuel deliveries completed at facilities with intact combustion systems. Automated dispatch system forensic investigation complete; system reconstituted with enhanced authorization verification before automated starts resume. Combustion turbine OEM contacted for hot section replacement components. Physical repair is underway at the kinetically attacked facilities. Grid operator managing the peaker gap through demand response, conservation voltage reduction, and import scheduling. Emissions permit status assessment for units with elevated operating hours.

Phase 3 — Medium-term (6 weeks to RT_HVLLC completion). Combustion turbine hot section replacement (8 to 14 weeks for standard frames). Sequential return to service as each unit completes repair. NERC documentation updates. Long-term assessment of whether the attacked peaker fleet, once repaired, remains sufficient for the governing winter emergency scenario or whether new capacity investment is required.

6.3 Critical Recovery Constraints

Combustion turbine OEM queue competition—the governing recovery constraint for Scenario A is when multiple units require the same hot section components simultaneously. Standard combustion turbine frames (GE Frame 7, Siemens SGT-800, and equivalent) have better component availability than coal steam turbines. Still, simultaneous demand from multiple units can create 4 to 8 week queue delays beyond normal lead times. Pre-established OEM priority agreements — specifying that RMR-designated peakers receive priority fabrication and delivery during declared grid emergencies — would reduce this constraint materially.

Emergency distillate delivery capacity. The New England distillate distribution system's surge capacity during peak heating season is the governing constraint for Scenario B recovery. The same terminal and trucking infrastructure serving residential heating oil during a cold snap may be unable to fulfill simultaneous emergency orders for peaker fuel delivery on the timelines required. Pre-positioned distillate storage at strategic locations — additional on-site tank capacity at the highest-consequence peaker facilities — is the most effective mitigation.

Automated dispatch reconstitution and authorization verification. Dispatch system forensic investigation and reconstitution must be completed before automated start capability is restored. The dispatch system must be re-validated with enhanced authorization verification — ensuring that every automated start requires a verified RTO dispatch signal or operator authorization — before returning to remote-only operation. Timeline: 3 to 5 weeks, consistent with other DBA-ES SCADA reconstitution timelines.

Emissions permit headroom. Units that have consumed permitted operating hours through spurious dispatches may require emergency permit modifications before returning to service. EPA's emergency permit modification process exists, but has not been exercised at the scale that a multi-facility Scenario B event would require. Pre-established emergency permit modification protocols for declared grid emergencies would reduce this constraint.

6.4 The Northeast Emergency Fuel Reserve

The compound consequence of Scenario C — simultaneous gas supply curtailment and oil-fired peaker loss — requires a dedicated recovery architecture element analogous to the Northeast Home Heating Oil Reserve addressed in DBA-OIL-FC3. No federal strategic reserve of distillate fuel for power generation exists; the NHOR holds heating oil for residential use, not generation fuel for grid emergency use. The distinction matters: heating oil and power generation distillate are chemically nearly identical (No. 2 fuel oil), but the delivery infrastructure — from the terminal to the power plant rather than to the residential tank — is different.

DOE, in coordination with ISO-NE and the Northeast state energy offices, should assess whether a strategic distillate reserve for New England grid emergency generation — pre-positioned at or near the highest-consequence RMR peaker facilities — would close the gap between the Scenario C consequence envelope and the available emergency supply response. The reserve volume required is the sum of the on-site storage replacement for the three to six highest-consequence RMR facilities, sized to sustain 72 hours of full-load emergency operation at each facility.

7. Gaps and Recommendations

7.1 Regulatory Gap: CIP Applicability Threshold Misalignment with Peaker Fleet Strategic Function

NERC CIP cybersecurity standards apply to generating facilities meeting the BES definition and rated above applicable size thresholds. Many oil-fired peakers — particularly older units below 100 MW or multi-unit sites whose individual unit ratings fall below CIP thresholds — may be exempt from CIP requirements while nonetheless holding RMR designations that establish their strategic importance to the grid. The CIP applicability threshold was not designed with the peaker fleet's emergency function in mind; a unit that runs 150 hours per year but is designated must-run for grid stability may be subject to less rigorous cybersecurity requirements than a much larger baseload unit that is easily replaceable in emergencies. NERC should evaluate applying CIP cybersecurity requirements to all RMR-designated oil-fired peakers regardless of size, using the RMR designation — rather than nameplate capacity — as the applicability trigger.

7.2 Automated Dispatch Authorization Verification

The Scenario B fuel depletion attack exploits the absence of a mandatory authorization verification requirement for automated start sequences at remotely operated peaker facilities. No NERC standard or RTO market rule currently requires that automated start events be corroborated against a verified dispatch authorization record before fuel consumption begins. NERC, working with the RTOs, should develop a standard requiring that automated dispatch systems at remotely operated generating facilities log every start event with a verified authorization identifier and that aggregate fuel consumption in any rolling 30-day period be compared against the verified dispatch log, with anomalous deviations triggering investigation. This is a low-cost, high-value control that directly addresses the Scenario B attack pathway.

7.3 Strategic Distillate Reserve for Northeast Grid Emergency Generation

No federal strategic reserve of distillate fuel exists for power generation use in New England grid emergencies. The assessment recommended in Section 6.4 should be treated as a formal gap requiring resolution before the next winter heating season, given the demonstrated vulnerability of the New England grid to compound gas-electric emergencies. The reserve volume, pre-positioning locations, and governance framework should be developed jointly by DOE, ISO-NE, and the Northeast state energy offices, using the Scenario C consequence envelope as the design basis.

7.4 Dual-Fuel Capability Verification

RMR designations for dual-fuel generating units are based on the unit's stated capability to operate on oil fuel. In practice, the oil-fuel systems of many nominally dual-fuel units have not been exercised in years — nozzles are clogged, fuel pump seals have degraded, and the startup procedures for cold oil-fuel starts have not been maintained in current operations manuals. An RMR unit that cannot actually start on oil fuel is a paper resource that exists in the ISO-NE capacity accreditation, but not in the real world. ISO-NE and NYISO should require annual oil-fuel start demonstrations — full cold starts on oil fuel under representative ambient conditions — for all RMR-designated dual-fuel units as a condition of maintaining the RMR designation and associated capacity payments.

7.5 Physical Security for Unmanned Remote Peaker Facilities

Oil-fired peakers that are unmanned or lightly staffed under normal operations have physical security postures calibrated for the consequences of unauthorized access to a low-utilization industrial site, not for the consequences of a precision kinetic attack on a must-run grid emergency resource. No regulatory framework requires physical security assessments for RMR-designated peaker facilities that fall below CIP-014 size thresholds. TSA, working with NERC and the RTOs, should develop a physical security guidance framework for RMR-designated peaker facilities — covering perimeter hardening, surveillance, and access control — that is proportional to the grid emergency consequence of the facility's loss rather than to its nameplate capacity or annual utilization rate.

Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial version. Establishes Design Basis Accident — Military Action scenarios for the oil-fired generation facility class. Issued under DBA-ES Electric Sector Framework v0.7. FC designation DBA-ES-GN-FC7. Applicability threshold: individual unit nameplate capacity at or above 100 MW burning distillate or residual fuel oil as primary fuel, or RMR designation regardless of size. Three scenarios were developed: Scenario A (pre-emptive destruction of peaker fleet before induced gas supply disruption — the backstop removal attack), Scenario B (fuel inventory depletion through spurious automated dispatch — invisible depletion via compromised automated start sequencing), Scenario C (Northeast peaker cluster attack during Winter Storm Uri analog — compound gas-electric winter emergency with cold-weather life-safety consequence). Dual Design Basis

			architecture established with three oil-fired-generation-specific AI governance interaction points. RT_HVLLC calibrated for combustion turbine hot sections: 8 to 14 weeks standard frames, 16 to 24 weeks older frames; steam unit subset approaching FC6 range. Northeast strategic distillate reserve gap identified. Regulatory gaps identified: CIP applicability threshold misalignment with peaker fleet strategic function, automated dispatch authorization verification, strategic distillate reserve, dual-fuel capability verification, and physical security for unmanned facilities.

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